Surrogate Inverse-Variance Weighting for Intervention Harvesting on Heavy-Tailed Click Logs

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Abstract

Under the *Position-Based Model* (PBM), the probability that a user clicks a document d shown for query q at position k factorises as $p_k\, r_{q,d}$, with $p_k \in (0,1]$ the position-only examination probability and $r_{q,d} \in [0,1]$ the query–document relevance. Intervention harvesting [1] estimates the position-bias ratio $p_k/p_{k'}$ from naturally occurring ranking variations by aggregating, across many (q,d) cells (indexed by i), per-cell click-rate ratios at positions k and k'. Each cell i contributes a per-cell weight $\omega_i \geq 0$; the choice of ω_i is decisive because impression counts N_k^i (the number of times cell i was shown at position k) are extremely heavy-tailed on real click logs: on the full KDD Cup 2012 Track 2 log, 60% of interventional (q, ad) pairs have ≤ 5 total impressions while 0.1% of pairs hold 42% of all impressions.

The intervention-harvesting estimator of Agarwal et al. uses *uniform* weighting ($\omega_i \equiv 1$); we propose two new count-only alternatives—the *min-count* weight $\omega_i = \min(N_k^i, N_{k'}^i)$ and the *harmonic* weight $\omega_i = N_k^i N_{k'}^i/(N_k^i + N_{k'}^i)$. *Why harmonic?* Under PBM, the per-impression click variance at cell i, position k is $p_k r_{q,d}(1-p_k r_{q,d})/N_k^i$ (Bernoulli variance), which depends on the unknown per-cell relevance. Replacing the relevance-dependent factor with a constant yields a *count-only surrogate* of this variance that is exact when the per-cell relevance is constant across cells, and a controlled approximation otherwise. Among count-only weights, harmonic is the inverse-variance optimum on this surrogate, and a Cauchy–Schwarz inequality guarantees it reduces the surrogate for any impression counts. The full inverse-variance optimum would require knowing $r_{q,d}$ per cell and is infeasible on heavy-tailed data.

We assess finite-sample efficiency at fixed weights via the closed-form delta-method asymptotic variance of the weighted ratio estimator [10, 12]. On the full KDD log, harmonic's asymptotic relative efficiency over uniform weighting is 4.5× and remains robust to weight capping. Under PBM mis-specification, different weight schemes target different weighted averages of cell-specific ratios—an estimand caveat we make explicit. Synthetic and Yahoo-LTR semi-synthetic experiments with ground truth corroborate the efficiency prediction (up to 54% variance reduction at high imbalance); min and harmonic are empirically comparable, and we recommend harmonic for its unconditional theoretical guarantee.

CIKM '26, Toronto, Canada

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/10.1145/nnnnnnn.nnnnnnn

ACM Reference Format:
Anonymous Author(s). 2026. Surrogate Inverse-Variance Weighting for Intervention Harvesting on Heavy-Tailed Click Logs. In *Proceedings of ACM CIKM (CIKM '26).* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Position bias systematically corrupts click signals used for learning-to-rank [7]; intervention harvesting [1] estimates position-bias ratios $R_{k,k'} = p_k/p_{k'}$ from naturally occurring ranking variations without explicit randomisation. On production click logs, the per-cell counts that feed this estimator are extraordinarily heavy-tailed. On the full KDD Cup 2012 Track 2 release (235M impressions; 4.6M interventional pairs at adjacent positions $(1,2)$), **60.7% of pairs have $N_1 + N_2 \leq 5$ while 0.1% of pairs carry 42% of all impressions** [8]. Public benchmarks can hide this shape: the released Baidu-ULTR [15] aggregation has 94% of (q, d, pos) cells with $N = 1$ and 78% of adjacent-pair impression ratios within $[0.5, 2]$, which we attribute to the per-cell deduplication used in the public release rather than to the underlying production logs (the unaggregated KDD release, by contrast, preserves the heavy-tailed production distribution).

Equal-weighting heteroscedastic estimators is statistically sub-optimal by classical inverse-variance theory [4], a result the meta-analysis literature has championed for decades. The intervention-harvesting estimator of [1] uses uniform per-cell weighting (each cell's click rate enters the sum with weight 1). **We propose two new count-aware alternatives**: the *min-count* weight $\omega_i = \min(N_k^i, N_{k'}^i)$ and the *harmonic* weight $\omega_i = N_k^i N_{k'}^i/(N_k^i + N_{k'}^i)$. Empirically, min and harmonic perform comparably on KDD (Section 4); the harmonic proposal is preferable because of its stronger theoretical foundation: it is the inverse-variance optimum of a relevance-agnostic surrogate (Theorem 2) and the only one of the two with an unconditional variance-reduction guarantee against uniform (Theorem 3). Existing variance-reduction arguments on heavy-tailed real logs have relied on *bootstrap* resampling, which we show can be misleading for inverse-variance-style weights on singleton-dominated data (Section 3).

Contributions. (1) We show that the harmonic weight is the inverse-variance optimum on a *relevance-agnostic count-only surrogate* of the per-cell variance under PBM (Theorem 2), and prove a Cauchy–Schwarz inequality showing it reduces this surrogate for any impression counts (Theorem 3). The exact inverse-variance oracle for the more general asymmetric surrogate is also characterised, and harmonic is its parameter-free special case (Theorem 4). **(2)** We adopt the textbook closed-form asymptotic variance of the weighted ratio estimator as a finite-sample diagnostic that is properly aligned with conditional-on-counts efficiency on heavy-tailed data, where naive pair-bootstrap is misleading. **(3)** On the full KDD log, harmonic delivers an asymptotic relative efficiency of 4.53 over uniform on this surrogate – meaning uniform weighting would need roughly 4.5× the data to match harmonic's precision – with the plug-in agreeing within 1–2% of empirical click-resample standard error and remaining robust to weight capping. A Cochran-Q heterogeneity test ($Q/\mathrm{df} = 3.83$) confirms substantial PBM mis-specification, under which the choice of weight is a precision/estimand trade-off rather than a search for a single "true" position-bias ratio. Synthetic and Yahoo-LTR experiments with ground truth corroborate.

2 Estimator and Theory

Notation. Throughout, q is a query, d a document/ad, k a result position, and i indexes the (q, d) cells in the interventional set. The Position-Based Model (PBM) assumes $P(\text{click} \mid q, d, k) = p_k r_{q,d}$ with $p_k \in (0, 1]$ the position-k examination probability and $r_{q,d} \in [0, 1]$ the (q, d) relevance; we abbreviate the per-cell relevance as r_i and write the true per-cell click probability as $\theta_k^i = p_k r_i$. We observe N_k^i impressions and C_k^i clicks for cell i at position k, with empirical rate $\hat{\theta}_k^i = C_k^i / N_k^i$. Our target is the position-bias ratio $R_{k,k'} = p_k / p_{k'}$. Per-cell weights are $\omega_i \geq 0$. Given the set of (q, d) pairs that appeared at both positions k and k', the estimator of [1] for $p_k / p_{k'}$ is

$$\hat{R}_{k,k'} = \frac{A}{B} = \frac{\sum_i \omega_i \hat{\theta}_k^i}{\sum_i \omega_i \hat{\theta}_{k'}^i}, \quad \hat{\theta}_k^i = \frac{C_k^i}{N_k^i}. \tag{1}$$

$\hat{R}_{k,k'}$ is the standard *self-normalised weighted ratio* estimator: a sum of weighted per-cell click rates divided by another such sum. We say a weight scheme is *ratio-unbiased* when, under PBM, the ratio of the conditional expectations of the numerator and denominator equals $p_k / p_{k'}$ – note this is not the same as $\mathbb{E}[\hat{R}] = p_k / p_{k'}$, because expectation does not commute with ratios; the finite-sample bias of $\hat{R}$ is $O(1/N)$ under standard regularity. Under PBM, every count-only weight $\omega_i = \omega(N_k^i, N_{k'}^i)$ satisfies this property, and $\hat{R}$ is consistent for $p_k / p_{k'}$ [1].

Proposition 1 (Conditional delta-method variance). *Given the vector of impression counts* $\mathbf{N} = \{N_k^i\}$, *the delta method [12] gives*

$$\mathrm{Var}(\hat{R}_{k,k'} \mid \mathbf{N}) \approx R_{k,k'}^2 \left[\alpha V_k(\omega) + \beta V_{k'}(\omega)\right],$$

with $V_r(\omega) = \sum_i \omega_i^2 / N_r^i / (\sum_i \omega_i)^2$, $\alpha = (1 - p_k)/p_k$, *and* $\beta = (1 - p_{k'})/p_{k'}$.

Relevance-agnostic count-only IVW.. The exact Bernoulli variance is $\mathrm{Var}(\hat{\theta}_k^i) = p_k r_i (1 - p_k r_i)/N_k^i$, which depends on the cell relevance r_i. Plug-in inverse-variance weights with $\hat{r}_i$ estimated per cell define the *θ-aware oracle* of Theorem 4; in heavy-tailed click logs $\hat{r}_i$ is itself extremely noisy on the ∼60% singleton mass, making this oracle infeasible in practice (Table 2: ARE = 0.002). We therefore work with a *relevance-agnostic surrogate* that drops the r_i-dependent factor from the cell-level variance; this surrogate is exact when the per-cell relevance r_i is constant across cells, and a controlled approximation otherwise. The inverse-variance optimum [4] on this surrogate, restricted to count-only weights, is the harmonic mean of the two cell counts. We make this precise:

Theorem 2 (Count-only IVW on the surrogate). *Among all non-negative weights* $\omega_i = \omega(N_k^i, N_{k'}^i)$, *harmonic weighting minimises the count-only delta-method variance surrogate* $V_k(\omega) + V_{k'}(\omega)$.

Theorem 3 (Surrogate guarantee, unconditional in counts). *For any impression counts* $\{N_k^i, N_{k'}^i\}_{i=1}^n$, $V_k(\omega^h) + V_{k'}(\omega^h) \leq V_k(\mathbf{1}) + V_{k'}(\mathbf{1})$. *The guarantee is unconditional in the counts but conditional on* $V_k + V_{k'}$ *as the objective; it implies improved* $\mathrm{Var}(\hat{R} \mid \mathbf{N})$ *via Proposition 1 when* $\alpha \approx \beta$. *For the asymmetric* $\alpha V_k + \beta V_{k'}$ *surrogate the bound is empirical only, but we verify on full KDD that harmonic reduces it by essentially the same factor as the symmetric one (Section 4).*

Sketch. Substituting $\omega_i^h = N_k^i N_{k'}^i / (N_k^i + N_{k'}^i)$ collapses the LHS to $1/W$ with $W = \sum_i \omega_i^h$. The RHS is $(1/n^2) \sum_i 1/\omega_i^h$. Cauchy–Schwarz gives $n^2 \leq W \sum_i 1/\omega_i^h$. □

Theorem 4 (Exact asymmetric oracle). $\alpha V_k(\omega) + \beta V_{k'}(\omega) \geq (\sum_i 1/s_i)^{-1}$ *with* $s_i = \alpha / N_k^i + \beta / N_{k'}^i$, *attained at* $\omega_i^\star \propto N_k^i N_{k'}^i / (\alpha N_{k'}^i + \beta N_k^i)$. *Harmonic is the* $\alpha = \beta$ *special case.*

Theorem 3 is unconditional in the counts – a property that distinguishes harmonic from the min-count variant, which has no such guarantee. Theorem 4 provides the oracle benchmark; in synthetic experiments substituting the true α, β into $\omega^\star$ yields variance within ±2% of harmonic. Complete proofs and additional sensitivity analyses are provided in the supplementary material.

Limits of the symmetric guarantee. The unconditional bound in Theorem 3 concerns $V_k + V_{k'}$. The asymmetric analogue $\alpha V_k(\omega^h) + \beta V_{k'}(\omega^h) \leq \alpha V_k(\mathbf{1}) + \beta V_{k'}(\mathbf{1})$ does *not* hold unconditionally: a closed-form $n = 2$ counterexample with $(N_k^1, N_{k'}^1) = (1, 100)$, $(N_k^2, N_{k'}^2) = (100, 1)$ and $(\alpha, \beta) = (0.01, 1)$ has harmonic strictly worse than uniform on the asymmetric surrogate. The configurations under which this fails are degenerate (each cell informative at exactly one position); we have not encountered them in any realistic count distribution we tested. The $\omega^\star$ in Theorem 4 remains the exact oracle for all (α, β).

3 Plug-in Asymptotic Variance as Diagnostic

$\widetilde{V}$ *in closed form.* The delta method applied to $\hat{R}_{k,k'} = A/B$ gives

$$\widetilde{V}(\omega) = \frac{\sum_i \omega_i^2 \theta_k^i (1 - \theta_k^i)/N_k^i}{A^2} + \frac{\sum_i \omega_i^2 \theta_{k'}^i (1 - \theta_{k'}^i)/N_{k'}^i}{B^2}, \tag{2}$$

where $A = \sum_i \omega_i \theta_k^i$, $B = \sum_i \omega_i \theta_{k'}^i$, and clicks at the two positions are conditionally independent given counts. The standard error of $\hat{R}_{k,k'}$ is then $\mathrm{SE}(\hat{R}_{k,k'}) = |\hat{R}_{k,k'}| \sqrt{\widetilde{V}(\omega)}$. Plugging in shrinkage estimates $\hat{\theta}_k^i = (C_k^i + \bar{\theta}_k)/(N_k^i + 1)$ with $\bar{\theta}_k$ the marginal position-k CTR yields the textbook self-normalised IS variance estimator [9–11].

ARE.. For schemes a, b, $\mathrm{ARE}(a, b) = \widetilde{V}(\omega_b)/\widetilde{V}(\omega_a)$: scheme b needs $\mathrm{ARE}(a, b)$ times the data of a for equal precision. $\widetilde{V}$ scales as $1/\mathrm{ESS}(\omega)$, with Kong's effective sample size [9] $\mathrm{ESS}(\omega) = (\sum_i \omega_i)^2 / \sum_i \omega_i^2$.

Why pair-bootstrap is misaligned here. A naive bootstrap that resamples (q, d) pairs estimates a different quantity than the conditional-on-counts efficiency we care about, for three reasons. *(i)* Averaging over ∼ 60% near-singleton pairs under uniform produces a smooth resampling distribution whose tightness reflects

Table 1: Plug-in SE vs. click-resample SE across KDD subsamples, and ARE(harmonic, uniform) at each scale. Click resample only computed at small f for runtime; agreement is within 1–2% where measured. ARE grows with data volume because heavy-tail mass becomes more pronounced at larger scales.

f	pairs	$\widehat{\mathrm{SE}}_U^{\mathrm{plug}}$	$\widehat{\mathrm{SE}}_U^{\mathrm{boot}}$	$\widehat{\mathrm{SE}}_H^{\mathrm{plug}}$	ARE
10^{-3}	4.3K	0.131	0.132	0.111	1.66
10^{-2}	54K	0.039	0.038	0.028	2.21
10^{-1}	542K	0.012	–	0.007	3.07
1.0	4.61M	0.0042	–	0.0020	**4.53**

Table 2: Plug-in delta-method asymptotic variance $\widetilde{V}$, SE of $\hat{R}$, ARE vs. uniform, and ESS on full KDD Cup 2012 Track 2 ($n_{\mathrm{pairs}} = 4.61$M). Harmonic, min, and the count-only oracle all reach ARE ≈ 4.5. The full-θ oracle is pathological because per-cell $\hat{\theta}_i$ is too noisy at this scale.

Scheme	$\hat{R}$	SE	ARE	ESS
Uniform	1.683	4.24e-3	1.00	4.61M
Min	1.712	2.05e-3	4.41	2,526
Harmonic	1.710	2.02e-3	**4.53**	1,849
Count-oracle (0.5,1)	1.715	2.02e-3	4.57	1,760
θ-aware oracle	2.343	0.148	0.002	1.1

averaging-of-noise rather than estimator precision. *(ii)* Pair resampling perturbs the high-information cells that drive any inverse-variance estimator, so concentrated weight schemes are systematically penalised by the resampling design, not by their statistical behaviour. *(iii)* Pair-bootstrap entangles estimator efficiency with model fit; $\widetilde{V}$ separates them (Section 4). The appropriate empirical baseline against which to validate the plug-in is the click-resample (parametric Bernoulli bootstrap conditional on counts), which we adopt for calibration.

Calibration and stability at scale. Table 1 reports plug-in SE versus empirical click-resample SE across KDD sample fractions, together with ARE(harmonic, uniform) at each scale. Because both $\widetilde{V}$ and click-resample SE share the conditional Bernoulli + cross-cell-independence assumption, their agreement is primarily an implementation-correctness check; a stronger probe of the independence assumption would be a query-cluster bootstrap, which is sensitive to Cochran-Q-style heterogeneity. We report pair-bootstrap numbers in supplementary material. The ARE *grows* with sample size ($1.66 \longrightarrow 4.53$) because the heavy-tailed structure of the count distribution becomes more pronounced as more data is included rather than averaged out.

4 Real-Log Results: KDD Cup 2012 Track 2

Data. Full release: 149.6M raw rows expanding to 235.6M impressions, 63.8M unique $(q, \mathrm{ad}, \mathrm{pos})$ cells, 8.2M clicks (CTR 3.49%). Pair $(1, 2)$ interventional set: 4.61M (q, ad) pairs, 132.3M impressions, median pair size 4, max 2.47M; 60.7% of pairs have $N_1 + N_2 \leq 5$.

Table 3: Stratified ARE(harmonic, uniform) by $N_1 + N_2$ tertile on full KDD (the bottom quartile boundary collapses into the lowest tertile because singletons dominate). Schemes are tied on the noise-dominated strata; harmonic dominates $\sim 5\times$ on the high-information stratum where the actual estimation signal lives.

$N_1 + N_2$ tertile	n pairs	$\widetilde{V}_U$	ARE(h, u)
Lowest	1.99M	2.3e-5	1.02
Middle-low	1.38M	1.9e-5	1.09
Highest	1.25M	7.6e-6	**4.98**

Headline (Table 2). ARE(harmonic, uniform) = 4.53: **uniform requires $\sim 4.5\times$ the impressions of harmonic for matching precision** on the relevance-agnostic surrogate. The min-count proposal achieves 4.41 on the same surrogate (−2.7% relative): harmonic and min are essentially tied empirically on KDD, and we recommend harmonic because it carries the unconditional guarantee (Theorem 3) and the parameter-free derivation, not because of a large empirical margin. The count-only oracle (4.57) is the ceiling within the surrogate. The θ-aware oracle, which would be optimal under classical inverse-variance theory with known r_i, plugs in $\hat{r}_i$ that is too noisy on the singleton-dominated tail and collapses to ARE = 0.002: a substantive finding rather than a side note—it shows that count-only weighting is robust precisely where plug-in IVW fails.

Asymmetric surrogate (empirical). With $\hat{p}_1 = 0.0501$, $\hat{p}_2 = 0.0256$ estimated from the marginal CTRs, $\alpha = 18.97$, $\beta = 38.08$, $\alpha/\beta \approx 0.50$. Harmonic reduces $\alpha V_k + \beta V_{k'}$ by a factor of 4.75 versus uniform. Both this number and the symmetric reduction of 4.84 are deterministic functionals of the observed counts—the gap reflects the intrinsic asymmetry of (α, β), not Monte-Carlo noise—and the two are within 2% of each other. The closed-form asymmetric counterexample of Section 2 requires strongly *anti-correlated* $(N_k, N_{k'})$ across cells (each cell informative at exactly one position); on KDD the per-pair counts are positively correlated, as items popular at position 1 are typically also popular at position 2, and we conjecture this is generic for production logs.

Stratified ARE (Table 3). Schemes tie on noise singletons (where there is nothing to estimate) and harmonic dominates uniform by 5× on the high-information stratum. The gain comes from *not* treating singletons as equal to million-impression cells.

Weight-cap robustness. The harmonic ESS is 1,849 out of 4.61M pairs: concentration on a small subset of high-count cells is what produces the surrogate $\widetilde{V}$ reduction, and the trade-off is sensitivity to those cells, which we probe directly by capping. Capping ω_i^h at the 99.99th percentile (threshold 2,079; 462 pairs capped) lifts ESS to 5.2×10^4 and still gives ARE = 4.12; at the 99th percentile, ARE = 2.83 (ESS 7.6×10^5); even at the 90th, ARE = 1.86. Harmonic beats uniform robustly across capping levels.

Comparison of count-only weighting schemes. Table 4 compares harmonic to the natural alternatives on full KDD. Among count-only weights, harmonic is best on $\widetilde{V}$. Log-counting is too flat (treats

Table 4: Count-only weighting schemes compared on full KDD via plug-in $\widetilde{V}$. Schemes that concentrate more aggressively give lower $\widetilde{V}$ but smaller ESS. Harmonic is the inverse-variance optimum among count-only schemes (Theorem 2).

Scheme	$\hat{R}$	SE	ARE	ESS
Uniform	1.683	4.24e-3	1.00	4.61M
$\log(1+\min(N_k, N_{k'}))$	1.651	3.18e-3	1.71	3.17M
$\sqrt{N_k N_{k'}}$	1.706	2.28e-3	3.55	1,246
$\min(N_k, N_{k'})$	1.712	2.05e-3	4.41	2,526
Harmonic	1.710	2.02e-3	**4.53**	1,849
Harmonic, cap@99th pct	1.606	2.40e-3	2.83	765K

singletons nearly as well as million-impression cells); the geometric mean and min-count concentrate more aggressively but neither matches the inverse-variance optimum. Capping at the 99th percentile preserves a 2.83× advantage but shifts the estimand ($\hat{R}$: 1.71 → 1.61), consistent with the estimand discussion below.

Estimand under PBM mis-specification. PBM is a known approximation; on real logs different weight schemes estimate different functionals of cell-specific ratios. Specifically, the population limit of $\hat{R}$ under any weight ω is

$$R_\omega \;=\; \frac{\sum_i \omega_i\, \theta_k^i}{\sum_i \omega_i\, \theta_{k'}^i},$$

which equals $p_k/p_{k'}$ only when $\theta_k^i/\theta_{k'}^i$ is the same constant across cells—i.e. when PBM holds exactly. Under mis-specification each scheme thus targets a different weighted average of the cell-specific ratios $\rho_i = \theta_k^i/\theta_{k'}^i$ (uniform → 1.68, harmonic → 1.71, θ-aware oracle → 2.34 on KDD). The plug-in $\widetilde{V}$ comparison is a precision statement at fixed ω—a model-based efficiency comparison—and does *not* claim that one R_ω is the "correct" position-bias ratio. Choosing ω thus involves a precision/estimand trade-off; harmonic's attractive property is that, among count-only schemes, it concentrates on high-information cells (where ρ_i is estimated most precisely) without depending on noisy plug-in $\hat{\theta}_i$.

Quantifying PBM mis-specification (Cochran's Q). PBM is a known approximation, not a literal model – prior work [6, 14] has long established that real logs violate it. The relevant question is how *large* the deviation is. Cochran's Q on the 16,132 high-information pairs (≥ 5 clicks at each of positions 1 and 2) gives $Q = 6.17{\times}10^4$, df = 16,131, $Q/\text{df} = 3.83$: per-cell odds-ratio variability is ∼ 3.8× what binomial sampling alone would produce, well outside Poisson dispersion. This magnitude is the quantitative driver of the estimand drift discussed in Paragraph 4; plug-in $\widetilde{V}$ keeps the efficiency comparison well-posed by working at fixed ω.

5 Synthetic and Yahoo-LTR Validation

To validate the ARE prediction in settings with ground truth, we run two controlled experiments.

Anchor-item imbalance sweep (synthetic). Following [1]: M=10 positions, $p_k = k^{-\eta}$ with $\eta = 1$, AdjacentChain estimator. A fraction s of documents are "anchors" with heavy traffic at one position ($N \sim \text{Unif}(50\rho, 200\rho)$) and light traffic at the other ($N \sim \text{Unif}(1, 51)$); the remaining $1 - s$ are balanced ($N \sim \text{Unif}(25, 75)$ at both positions). Here s is the traffic split and ρ is the imbalance ratio (disjoint from r_i above); $s \in \{0.5, \ldots, 0.95\}$, $\rho \in \{1, \ldots, 100\}$, 500 trials per cell. At $s = 0.80$, $\rho = 100$ harmonic reduces variance by 43.6% vs. uniform and 5% vs. min; reduction grows to ∼ 54% at $s = 0.90, \rho = 50$. Across the full (s, ρ) grid the data-equivalence factor $\text{DEF}(s, \rho) = b_{\text{uni}}(s, \rho)/b_{\text{harm}}(s, \rho)$ – the ratio of base impressions per document at which each scheme first hits a fixed MSE threshold – has median 1.31, mean 1.30, max 1.94. Highly imbalanced cells achieve up to 2× data savings; near-balanced cells ($\rho = 1$) saturate at DEF ≈ 1.

Table 5: MSE decomposition under three explicit PBM violations (anchor imbalance 50:1, split 80/20, 5,000 trials). Count-only schemes share comparable bias^2 within each model; harmonic achieves the lowest variance, hence the lowest MSE.

Model	Scheme	MSE	Variance	Bias^2
Trust	Uniform	0.00487	0.00486	1.0e-5
	Min	0.00298	0.00297	2.0e-5
	Harmonic	**0.00272**	0.00272	0.0
Cascade	Uniform	0.01376	0.00276	0.01100
	Min	0.01242	0.00171	0.01071
	Harmonic	**0.01252**	0.00147	0.01105
Pos.-dep.	Uniform	0.04086	0.03973	0.00113
	Min	0.03896	0.03770	0.00127
	Harmonic	**0.03724**	0.03613	0.00111

Bias comparability under PBM violations. We additionally simulate three explicit PBM violation models – trust bias, cascade, and position-dependent relevance [1, 5] – at imbalance ratio 50:1, split 80/20, 5,000 trials. Across all three models the count-only schemes (uniform, min, harmonic) exhibit *comparable* bias^2, so harmonic's lower variance translates directly into lower MSE (Table 5). Under cascade, where history-dependent examination introduces a substantial structural mismatch (bias^2 ≈ 0.011 for every ratio-unbiased scheme), the variance edge is preserved but the practical MSE gain shrinks because bias dominates.

Yahoo-LTR semi-synthetic. Same simulator with graded relevance from the Yahoo Learning to Rank Challenge Set 1 test split (6,936 queries; relevance grades $g \in \{0, ..., 4\}$ mapped to $r(q, d) = (2^g - 1)/15$ following NDCG; click model $P(C = 1 \mid q, d, k) = p_k\, r(q, d)$; 200 trials). Harmonic reduces variance by ∼ 40% vs. uniform at imbalance $\rho = 50$ (Var $1.5{\times}10^{-4} \to 8.7{\times}10^{-5}$). With 200 trials, a delta-method approximation places the 95% CI on the variance-ratio at [0.49, 0.71], so the gain is well separated from 1.0 but not tightly pinned. Aggressive clipping trades variance for bias and yields ∼458× higher MSE.

6 Related Work

Intervention harvesting was introduced by [1]; [13] jointly estimates relevance and propensities via EM. Self-normalised weighted ratio estimators are standard in importance sampling and survey statistics [3, 9, 10]; inverse-variance weighting is canonical in meta-analysis [4]. The closest off-policy analogue is SNIPS [2, 11]. PBM violations on real logs are documented in [6, 14].

Why count-only PBM weighting is still worth studying. Given that PBM is approximate, EM-style joint estimation of relevance and propensities [13] is a natural alternative with its own bias–variance trade-offs (initialisation sensitivity, model choice). The two approaches are complementary: a relevance-agnostic weight is the right input to an EM M-step that re-uses count statistics. Empirical EM comparison is left to the regular-paper version.

7 Conclusion

Harmonic weighting is the inverse-variance optimum on a relevance-agnostic count-only surrogate of the PBM delta-method variance; the actual inverse-variance solution (θ-aware oracle) is plug-in infeasible at scale (ARE $=$ 0.002). On full KDD, $\mathrm{ARE}(\mathrm{harmonic}, \mathrm{uniform}) = 4.53$ on the surrogate, with our two proposals essentially tied empirically; harmonic is preferred because of Theorem 3's unconditional guarantee. Under PBM misspecification, choosing ω is a precision/estimand trade-off. Future work: shrinkage θ-aware oracle; EM [13] comparison; click models beyond PBM [6, 14].

GenAI Usage Disclosure

The authors used a large-language-model assistant for code generation (KDD aggregation, plug-in variance script, LaTeX table formatting), and for copy-editing. All experimental design, mathematical derivations (Theorems 2–4 and Eq. (2)), interpretation, citations, and final wording were authored and verified by the authors. No LLM-generated content was used without human review. Full code, scripts, and data-processing pipelines are provided as supplementary material.

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